07

North South University

Department of Electrical & Computer Engineering

Lab Report

Experiment No: 07

Experiment Title: Build a single cycle datapath

Course Code: CSE332L

Section: 07

Course Name: Computer Organization & Architecture Lab

Lab Group #: 02

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Objective:

1. Design an instruction fetch unit of the datapath.

2. Design an R-formal datapath.

Equipment:

- Logisim tool.

Block Diagram:

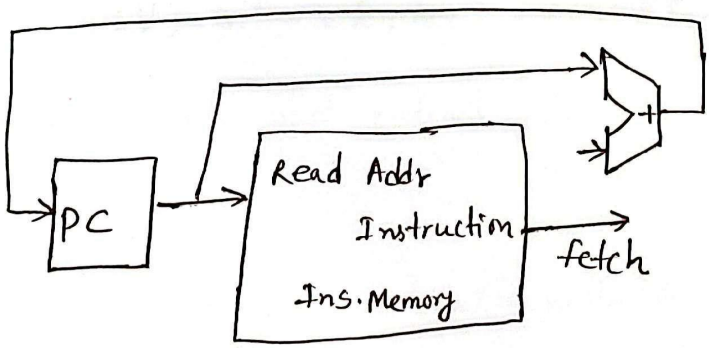


Fig: Instruction fetch unit.

Discussion:

In this experiment, we designed an instruction ~~fetch~~ unit. ~~and~~ the R-type datapath. ~~Our~~

Our ISA was 16 bit. To build the ISA we needed, register, adder and rom. we also used clock and clear for the register. Design of Instruction fetch unit was very simple.

In the next part of the experiment we designed R-type data path. To complete the datapath we used our previously designed ISA, 16 bit register file, 16 bit ALU, bit extender. Moreover, we connected a clock signal and output pin to show the output.

Designing the R-type datapath was a bit tricky but finaly we were able to complete the task. there was no hardware related issues.